# Gait Deviations Associated With Concussion: A Systematic Review

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**Abstract**

**Background:** Gait deviations resulting from concussion are important to consider in the diagnosis, treatment progression, and return to activity after a concussion. **Objective:** To identify quantifiable gait deviations associated with concussion across populations and time since injury. **Methods and Materials:** Six electronic databases were systematically searched from January 1974 to September 2016. Studies selected included original data, had an analytic design, and reported a quantifiable gait parameter in individuals who had sustained a concussion as defined by the American Congress of Rehabilitation Medicine or related definitions. Preferred Reporting Items for Systematic reviews and Meta-Analysis guidelines were followed. Two independent authors assessed study quality [Downs and Black (DB) criteria] and level of evidence (Oxford Center of Evidence-Based Medicine Model). **Results:** Of 2650 potentially relevant articles, 21 level 4 studies were included. The median DB score was 12/33 (range 10-16). Heterogeneity in gait parameters and timing of postconcussion testing precluded meta-analysis. There is consistent level 4 evidence of increased medial-lateral center-of-mass displacement, and inconsistent level 4 evidence of decreased gait velocity after concussion. Further, there is preliminary level 4 evidence that gait deficits may exist beyond the typical 10-day recovery period and return to activity. **Conclusion:** These findings suggest that individuals who have suffered a concussion may sway more in the frontal plane, and walk slower compared to healthy controls. Consensus about the most important gait parameters for concussion diagnosis and clinical management are lacking. Further, high-quality prospective cohort studies evaluating changes in gait from time of concussion to return to activity, sport, recreation and/or work are needed.

**Key Words:** concussion, gait, mild traumatic brain injury

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## INTRODUCTION

Concussion is the most frequent subtype of mild traumatic brain injury (mTBI).[1] The World Health Organization's (WHO) Collaborating Centre Task Force on mTBI estimates that the annual incidence of concussion is 100 to 300/100,000 emergency department visits.[1] However, as many concussions go unreported, it is estimated that the true annual incidence of concussion may be closer to 600/100 000.[1] Concussion is most prevalent in adolescents and young adult males, and is commonly attributed to sport participation, motor vehicle collision, and/or falls.[1]

A concussion is defined as a sequela of pathophysiological events that affect the brain as a result of a direct or indirect trauma to the head.[2] Concussions are not associated with alterations on standard structural imaging but can result in a vast range of acute neurophysiological signs and a multitude of symptoms that may persist for differing lengths of time.[2] Common short-term signs and symptoms include physical (eg, imbalance, loss of consciousness, or gait unsteadiness) and neurocognitive (eg, effected memory or reaction time) manifestations, sleep disturbance, and behavioral changes (eg, irritability).[2] Possible long-term consequences of concussion include cognitive impairments,[3] altered postural control,[4] gait impairment,[5] and increased risk of musculoskeletal injury.[6]

Concussion diagnosis is multidimensional and involves the assessment of physical (eg, loss of consciousness) and somatic symptoms (eg, headache), cognition (eg, reaction time), sleep quality, behavior (eg, irritability), and balance function.[2] The American Congress of Rehabilitation Medicine (ACRM) has defined clinical criteria that are widely accepted and used in the field of neurophysiology and rehabilitation to diagnose concussion.[7] Similar guidelines have been developed by the American Academy of Neurology (AAN),[7] the Consensus Statement on Concussion in Sport (CISG),[2] and the National Athletic Training Association (NATA; Table 1).[8]

One functional task that is commonly used to assist in concussion diagnosis, treatment progression, or return to activity, sport, recreation, and/or work decisions is gait. Gait is defined as "a method of locomotion involving the use of the 2 legs, alternately, to provide both support and propulsion."[9] Gait can be evaluated using self-report, qualitative (eg, rating of functional compensations, asymmetries, impairments or efficiency) or quantitative (eg, objective measurement of gait with tools such as motion analysis) methods.[10] Self-report and qualitative evaluation techniques are inherently subjective and

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**TABLE 1. Summary of Concussion Definitions and Diagnosic Guidelines**

| Organization | Concussion Definition and Diagnosis |
|---|---|
| AAN | Definition: A clinical syndrome of biomechanically induced alteration of brain function, typically affecting memory and orientation, which may involve LOC. |
| | Diagnosis based on |
| | Post-Concussion Symptom Scale or Graded Symptom Checklist. |
| | The Standardized Assessment of Concussion (SAC) |
| | Neuropsychological testing |
| | The Balance Error Scoring System |
| | The Sensory Organization Test |
| ACRM | Definition: A traumatically induced physiological disruption of brain function. |
| | Diagnosis based on |
| | Loss of consciousness (LOC) |
| | Loss of memory of immediately before or after an incident |
| | Alteration in mental state at the time of an incident (ie, dazed, disoriented, or confused) |
| | Focal neurologic deficits) where the injury severity does not exceed: LOC of $\sim \leq$30 min: Glasgow Coma Scale (GCS) score of 13–15 at 30 min; Posttraumatic amnesia (PTA) not $>$24 h. |
| CIGS | Definition: A brain injury and is defined as a complex pathophysiological process affecting the brain, induced by biomechanical forces. It may be caused either by a direct blow to the head, face, neck or elsewhere on the body with an "impulsive" force transmitted to the head, and typically results in the rapid onset of short-lived impairment of neurologic function that resolves spontaneously. However, in some cases, symptoms and signs may evolve over a number of minutes to hours. Further, concussion may result in neuropathological changes, but the acute clinical symptoms largely reflect a functional disturbance rather than a structural injury and, as such, no abnormality is seen on standard structural neuroimaging studies. Concussion results in a graded set of clinical symptoms that may or may not involve LOC. Resolution of the clinical and cognitive symptoms typically follows a sequential course. However, it is important to note that in some cases symptoms may be prolonged. |
| | Diagnosis based on |
| | Symptoms—somatic (ie, headache), cognitive (ie, feeling like in a fog) and/or emotional symptoms (ie, ability) |
| | Physical signs (ie, loss of consciousness, PTA) |
| | Behavioral changes (ie, irritability) |
| | Cognitive impairment (ie, slowed reaction times) |
| | Sleep disturbance (ie, insomnia) |
| NATA | Definition: Trauma-induced alteration in mental status that may or may not involve LOC. |
| | Diagnosis based on |
| | Clinical evaluation supported by assessment tools. |
| | A brief concussion-evaluation tool (ie, SAC) in conjunction with symptom and motor-control evaluation for rapid assessment. |

may result in inaccurate diagnosis, treatment, or return to activity, sport, recreation, and/or work decisions.[10] Conversely, quantitative gait assessment techniques enable the consideration of a variety of parameters and provide a more robust and reliable basis for diagnosis and decision making.[10]

To date, most research examining the influence of concussion on gait has employed laboratory motion capture analysis systems to examine differences in kinematics and kinetics between individuals who have suffered a concussion and those who have not.[11,12] These investigations have used single (eg, gait alone) and/or dual (eg, gait while conducting a mental task, avoiding obstacles, and/or responding to auditory stimuli) tasks and have been conducted at various intervals (eg, 1, 2, 4, 8 weeks) postconcussion.[11,12]

Although gait is commonly considered in the clinical assessment of concussion,[13] there is a lack of consensus regarding which gait parameters are the most important for concussion diagnosis or to inform return to activity, sport, recreation and/or work decisions. The primary objective of this systematic review is to identify quantifiable gait deviations associated with concussion based on time since injury. The findings of this review will inform future research aimed at identifying which gait parameters are the most important to consider for clinical diagnosis of concussion and return to activity, sport, recreation and/or work decisions.

## METHODS

This review was registered in the PROSPERO database (CRD 42016032529) and conducted according to the Preferred Reporting Items for Systematic reviews and Meta- Analysis guidelines.[14]



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### Data Sources and Search

Six electronic databases [MEDLINE, CINAHL (Cumulative Index to Nursing and Allied Health Literature), Sport Discus, SCOPUS, EMBASE (Excerpta medical database) and PsycINFO] were searched from January 1, 1974 (Glasgow Coma Scale development)[15] to September 29, 2016 to identify relevant studies. The combination of medical subject headings and text words used to execute the literature search was developed in consultation with a librarian scientist (LD). Appendix 1 outlines the search term combinations for each database. Limits included; English or Arabic language, human participants and analytical concussion studies published in peer-reviewed journals. Returned articles were organized using the reference management software package RefWorks (ProQuest, 2016). The number of articles obtained from each search strategy by database was recorded and a running total constructed. After removing duplicate articles, the titles and corresponding abstracts of all returned articles were independently reviewed by 2 of the authors (T.S.M. and D.P.G. or J.L.W.) blinded to journal title, and author(s) using a Microsoft Excel workbook designed specifically for screening.[16] Data were compiled and consensus on items for which there was disagreement was reached through rater discussion. Before the title and abstract reviews, all authors independently screened a random sample of 120 titles and abstracts and reached a strong agreement with the lead author (probability of random agreement = 0.82, Cohen's Kappa = 0.91) using the same excel workbook.[16] Finally, 2 authors (T.S.M. and D.P.G. or J.L.W.) reviewed the full text of all potentially relevant studies to determine final study selection. Disagreements were resolved via author consensus. During the full text review, the reference lists of all potentially relevant articles were hand-searched to identify any potentially relevant studies that had not been identified in the database search.

### Study Selection

Studies were included if they represented primary research published in peer-reviewed journals, analytical study design, and contained original data that investigated the association between a quantifiable gait parameter (eg, step, stride, center of mass parameters) and concussion as defined according to one of the AAN, ACRM, CISG, or NATA criteria (Table 1). Studies were excluded if they involved: participants with moderate or severe TBI (eg, involved brain hemorrhage, skull fracture, neuroimaging abnormality, or open head injury), or brain pathology (eg, cardiovascular accident/stroke), animal models or cadavers. Further, reviews articles, meta-analyses, case studies, case series, editorials, commentaries, opinion-based papers and conference proceedings were excluded.

### Data Extraction and Study Rating Process

Data extracted from each study included: study year, design (repeated or nonrepeated measures), location, population (age, sex, athletic vs nonathletic, symptomatic vs nonsymptomatic, and sample size); concussion definition; gait parameters assessed; tasks performed; timing of postconcussion testing (hours, days, weeks, years, or a point after return to physical activity or play) and results (point estimates and measures of variability) where available. The first author (T.S.M.) performed the initial data extraction, and data accuracy was ensured by D.P.G. and J.L.W. Data extraction disagreements were resolved by author consensus.

Two authors (T.S.M. and D.P.G. or J.L.W.) independently evaluated the quality and level of evidence of each study. Quality of evidence was assessed based on criteria for internal validity (study design, quality of reporting, presence of selection and misclassification bias, potential confounding) and external validity (generalizability) using the Downs and Black (DB) quality assessment tool.[17] This tool assigns a score calculated out of 32 points (11 points for reporting, 3 points for external validity, 7 points for bias, 6 points for confounding and 5 for power) for each study. The level of evidence represented for each study was categorized according to the Oxford Centre of Evidence Based Medicine (OCEBM) model.[18] Disagreements in DB and OCEBM rating were resolved by rater consensus.

### Data Synthesis

The quantity, quality, and level of evidence for the most commonly investigated gait parameters across 3 time periods (ie, less than or equal to the typical 10-day postconcussion recovery period, greater than the typical 10-day postconcussion recovery period and after return to any activity including physical activity, sport and work) were collated. A cut-point of 10 days postconcussion was used as it has previously been reported that 80% to 90% of concussed individuals recover within 7 to 10-day postinjury.[2]

## RESULTS

### Identification of Studies

An overview of the study identification process is shown in Figure 1 and Appendix 2 summarizes the identification of unique articles by database. The initial search yielded 2650 articles, 1187 duplicates were removed leaving 1463 potentially relevant articles. After the removal of studies not meeting inclusion criteria based on title and abstract reviews, this number was reduced to 92. One hundred and thirty-nine studies were excluded on study design (9 case studies, 15 case reports, 10 case series, 8 commentaries, 7 editorials, 7 book chapters, 7 abstracts, 42 reviews, 2 meta-analysis, 31 conference proceedings, 1 survey), 721 did not fit the criteria for concussion, 3 only reported subjective outcomes, 265 did not investigate the association between concussion and gait, and 243 involved animal models. Subsequent to full article screening, 71 studies were excluded leaving 21 studies deemed appropriate for inclusion to the systematic review. No additional articles were identified through reference list searches. Meta-analyses were precluded due to the heterogeneity of investigated gait parameters and timing of postconcussion testing (see columns 4 and 5 in Table 2).

### Study Characteristics

Characteristics of the 21 included studies are summarized in Table 2.[13,19–38] The 21 studies were published between 2005 and 2016, and 16 (76%) were performed by 3 research teams.[11,13,21–24,28,30–38] Nineteen of the 21 studies were conducted in the US, with the remaining 2 taking place in Canada and Norway.[25,29] Six of the 21 studies were cross-sectional in nature[19,27,29,34,35,38] while the remaining



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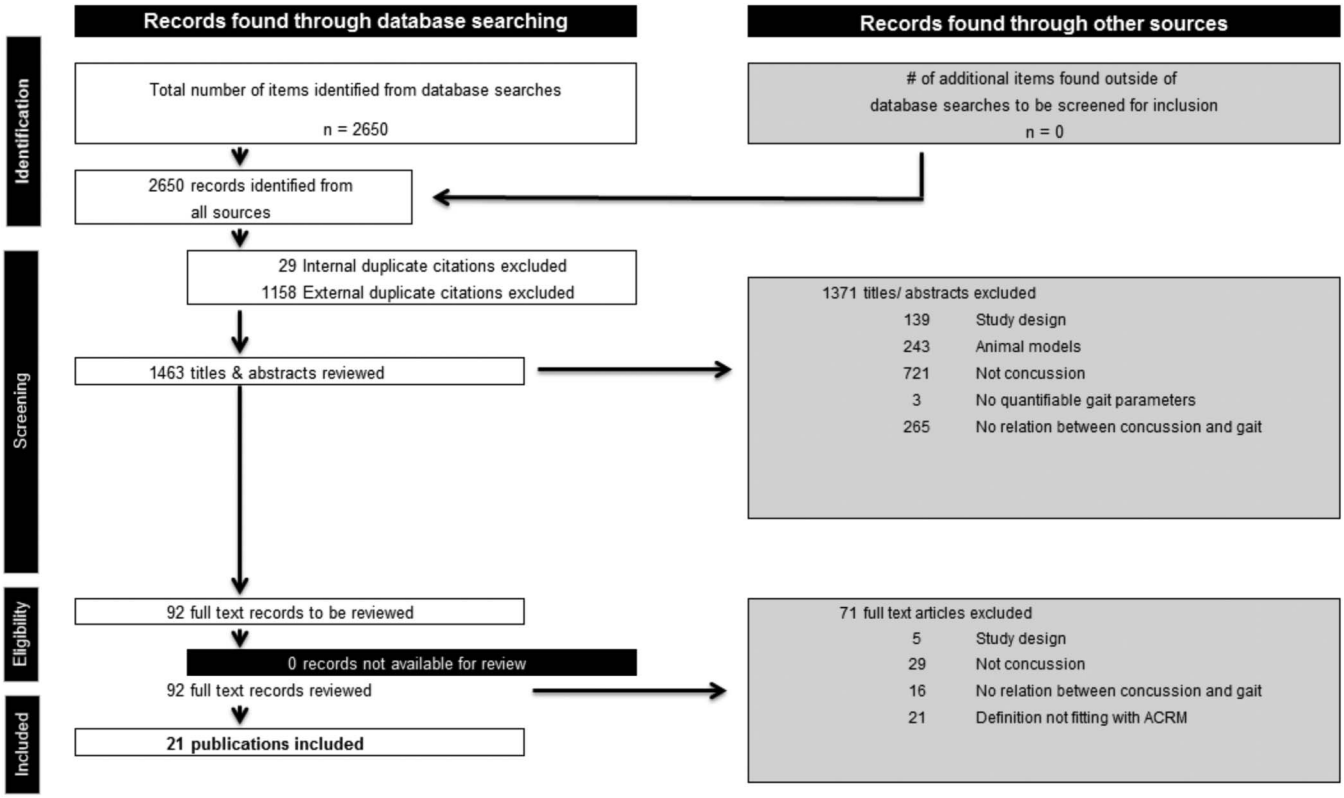


**Figure 1.**
Study identification PRISMA flowsheet.

incorporated repeated measures. The total number of participants assessed across studies was 1120. Participants ranged in age from 5 to 53 years and included 675 males and 342 females (2 studies did not report participant sex). Fourteen (66%) of the studies investigated gait parameters in athletes,[19–23,25–27,31–33,35,37,38] 4 (19%) considered high school and/or college students described as being involved in athletic-like activities,[24,28,34,35] 2 considered adults,[29,30] and 1 did not describe the source population.[13] Concussion was defined according to the AAN criteria in 11 studies (52%),[25,27,30–38] the CISG criteria in 9 studies (43%),[13,19–24,26,28] and the ACRM criteria in 1 study.[29] None of the included studies provided sufficient information to determine if participants were symptomatic or if they had been medically cleared to return to sport at the time of testing.

A variety of different gait tasks were used across the included studies. The majority (18/21) of studies used either a single[19–21,23–31,33–38] and/or a dual gait task (17/21),[13,19,21–24,27,28,30–38] while 4 used an obstacle crossing task.[27,31,32,35] The most commonly investigated gait parameters included gait velocity (GV; 15 studies),[13,19,20,24–29,33–38] center-of-mass displacement (COMD; 13 studies),[21–24,28,30,31,33–38] and center-of-mass velocity (COMV; 13 studies).[21–24,28,30,31,33–38] Other gait parameters assessed included center-of-mass, center-of-pressure separation (COM-COPS),[30,31,33–38] step width (STW),[25,28,34,35,37,38] stride length (SL),[19,34,35,37,38] stride time (SRT),[34,35,37,38] step length (STL),[25,28] propulsive and breaking forces percentage,[26] trunk roll angle and swing time,[25] lateral dynamic stability margin,[25] and obstacle-foot clearance.[32] Most (19/21) of the included studies gathered kinematic and kinetic gait parameters using a traditional motion analysis system while 2 of the studies employed wearable technology (gait analysis sensors).[13,19] The median interval between concussion and gait assessment across studies was 14 days (range 24 hours to 4 years). Sixteen studies reported gait deviations within the typical 10-day postconcussion recovery period,[13,19,20,22,24,26–28,30,31,33–38] 10 reported deviations after the typical 10-day postconcussion recovery period,[13,22–24,28,30,31,33,36,37] and 3 reported deviations after return to activity.[21,23,29] The majority (18/21) of the included studies compared concussed individuals and match-healthy controls, while 3 studies,[20,21,29] used baseline measures for comparison. Of the 21 included studies, 2 reported the clinometric properties of the measurement system employed in their investigation,[19,29] and 2 studies reported the diagnostic accuracy (sensitivity and/or specificity) of gait parameters (ie, GV and COM deviations) to differentiate between individuals who have suffered a concussion and those who have not.[13,20]

### *Study Quality and Level of Evidence*

The highest level of evidence demonstrated by the included studies as per the OCEMB levels of evidence model was level 4, corresponding to cross-sectional or case control studies, or poor quality prognostic cohort study.

Based on the DB criteria, the median methodological quality rating for the included studies was 12/33 (range 10-16). The DB is designed to evaluate the methodological quality of a scientific study and can be applied to both interventional and observational studies. As all of the included studies were observational in nature, 7 items (4, 8, 14, 19, 23, 24, and 27) totaling 10 points on the DB checklist were not applicable. The most consistent methodological weaknesses of the included



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**TABLE 2. Summary of Data Extraction**

| Study (Year, Design Follow-Up, Country) | Participants (Description, Age, Sex, Symptoms, Sample Size) | Concussion Definition | Gait Outcome(s), Time Point(s), and Task(s) | Gait Outcome(s) Reported as Significantly Different With Concussion | Gait Outcome(s) Reported as Not Significantly Different With Concussion | DB |
|---|---|---|---|---|---|---|
| Howell et al[19] | Adolescent athletes | CISG[2] | Outcome(s): 1. Average gait speed (m/s); 2. Cadence (steps/min); 3. Double support time (% gait cycle); 4. Gait cycle duration (s); 5. Stride length (m) | 5. Smaller stride length (≥2 concussions) walk + cognitive task ($P = 0.010$, $d = 0.070$) | 1. Average gait speed; 2. Cadence; 3. Double support time; 4. Gait cycle duration | 12 |
| No repeated measures | Mean age (range): CONC: 16 (14-17) >1 CONC: 17 (15-18); CONT: 15 (14-16) | | Time point(s): Within 14 d of concussion | | | |
| USA | Female: 68 | | Task(s): 1. Walking (self-selected speed); 2. Walk + cognitive task (spelling, counting, or reciting backwards) | | | |
| | n = 68 (37 concussions) | | | | | |
| | | | | | | |
| Galetta et al[20] | Youth and collegiate athletes | CISG[2] | Outcome(s): 1. Timed Tandem gait(s) | 1. Increased Tandem gait time ($P = 0.020$) | None | 10 |
| Repeated measures | Mean age (SD; range): CONC: Youth: 11 (3; 5-17); Collegiate: 20 (1; 18-23); CONT: Baseline data | | Time point(s): 1. Pre-season; 2. At concussion | | | |
| USA | Female: 61; Male: 271 | | Task(s): 1. Walking as fast as possible | | | |
| | n = 332 (12 concussions) | | | | | |
| | | | | | | |
| Howell et al[13] | Not reported | CISG[2] | Outcome(s): 1. GV (m/s); 2. Peak A–P acceleration (m/s$^2$); 3. Peak M–L acceleration (m/s$^2$) | 1. Decreased GV at 72 h ($P = 0.003$), 1 wk ($P = 0.013$), 2 wks ($P = 0.031$); 3. Decreased peak M–L acceleration at weeks 1, 2, 4, 8 ($F_{1,14} = 5.770$, $P = 0.040$) | 2. Peak A–P acceleration | 12 |
| Repeated measures | Mean age (SD; range); CONT 20: (4.5; 15-24); CONC: 19 (5.5; 14-25) | | Time point(s): 72 h, 1, 2, 4, 8 wks postconcussion | | | |
| USA | Female: 7; Male: 10 | | Task(s): 1. Walk + MAS | | | |
| | n = 17 (10 concussions) | | | | | |
| | | | | | | |
| Howell et al[21] | High school and collegiate athletes | CISG[2] | Outcome(s): 1. Total M–L COMD (cm); 2. Peak linear M–L COMV (cm/s); 3. Peak anterior COMV (cm/s) | 1. Increased total M–L COMD during Walk + MAS ($P = 0.004$); 2. Increased peak M–L COMV during Walk + MAS ($P = 0.048$) | 3. Peak anterior COMV | 11 |
| Repeated measures | Mean age (SD; range): CONC: 17 (2.9, 14-25); CONT: Baseline data | | Time point(s): within 2 mo after return to physical activities | | | |
| USA | Female: 8; Male: 21 | | Task(s): 1. Walk; 2. Walk + MAS | | | |
| | n = 29 (29 concussions) | | | | | |
| | | | | | | |



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**TABLE 2. Summary of Data Extraction** *(Continued)*

| Study (Year, Design Follow-Up, Country) | Participants (Description, Age, Sex, Symptoms, Sample Size) | Concussion Definition | Gait Outcome(s), Time Point(s), and Task(s) | Gait Outcome(s) Reported as Significantly Different With Concussion | Gait Outcome(s) Reported as Not Significantly Different With Concussion | DB |
|---|---|---|---|---|---|---|
| Howell et al[22] | High school & collegiate athletes. | CISG[2] | Outcome(s): 1. Total M–L COMD (m); 2. Anterior COMV (m/s); 3. M–L COMV (m/s) | 1. Increased total M–L COMD at 72 h ($F_{3,75} = 4.770$, $P = 0.004$), and 1, 2, 4, 8 wks ($F_{3,75} = 5.260$, $P = 0.002$); 2. Decreased peak anterior COMV at 72 h ($F_{3,75} = 8.360$, $P = 0.001$); 3. Increased peak M–L COMV at 72 h ($F_{3,75} = 5.760$, $P < 0.001$), across 8 wks ($F_{12,284} = 2.200$, $P = 0.012$) and at 8 wks ($P = 0.004$) | None | 16 |
| Repeated measures | Mean age (SD; range): CONC (adult): 20 (2.4; 18-27); CONT (adolescent): 15 (1.1; 14-17); CON (adult): 20 (2.1; 18-26); CONT (adolescent): 16 (1.1; 14-17) | | Time point(s): 72 h; 1, 2, 8 wks postconcussion | | | |
| USA | Female: 24; Male: 52 | | Task(s): 1. Walk + MAS | | | |
| | n = 76 (38 concussions) | | | | | |
| | | | | | | |
| Howell et al[23] | Athletes | CISG[2] | Outcome(s): 1. M–L COMD; 2. Peak M–L COMV; 3. Peak anterior COMV | 1. Increased M–L COMD during Walk + MAS ($F_{1,36} = 7.65$, $P = 0.002$) | 2. Peak M–L COMV; 3. Peak anterior COMV | 13 |
| Repeated measures | Mean age (SD; range): CONT: 16 (1.1; 15-17); CONC: 15 (1.4; 14-17) | | Time point(s): 72 h; 1, 2, 4, 8 wks; Pre and post return to physical activities (within 2 mo) | | | |
| USA | Female: 6; Male: 32 | | Task(s): 1. Walk; 2. Walk + MAS | | | |
| | n = 38 (19 concussions) | | | | | |
| | | | | | | |
| Howell et al[24] | High school students | CISG[2] | Outcome(s): 1. Total M–L COMD (m); 2. Peak anterior COMV (m/s); 3. Peak M–L COMV (m/s); 4. Average GV (m/s) | 1. Increased total M–L COMD at 72 h, 1, 2, 4, 8 wks during Walk + MAS and Walk + Q & A ($F_{3,129} = 5.310$, $P = 0.004$) | 2. Peak anterior COMV; 3. Peak M–L COMV; 4. Average GV | 11 |
| Repeated measures | Mean age (SD; range): CONT: 16 (1.3; 14-17); CONC: 15 (1.3; 14-17) | | Time point(s): 72 h; 1, 2, 4, 8 wks postconcussion | | | |
| USA | Female: 6; Male: 40 | | Task(s): 1. Walk; 2. Walk + SAS; 3. Walk + MAS; 4. Walk + Q & A | | | |
| | n = 46 (23 concussions) | | | | | |
| | | | | | | |
| Powers et al[25] | Collegiate athletes | AAN[7] | Outcome(s): 1. Step length (cm); 2. Step width (cm); 3. GV (cm/s); 4. Trunk roll angle (degrees); 5. Trunk swing time (s); 6. Minimum lateral dynamic stability margin (cm) | 5. Increased swing time at acute phase ($P = 0.009$, Cohen's $d = 1.400$), at return to practice phase ($P = 0.005$, Cohen's $d = 1.520$); 6. Increased lateral dynamic stability margin at acute phase ($P = 0.027$, Cohen's $d = 1.150$) | 1. Step length; 2. Step width; 3. GV; 4. Trunk roll angle | 15 |



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**TABLE 2. Summary of Data Extraction** *(Continued)*

| Study (Year, Design Follow-Up, Country) | Participants (Description, Age, Sex, Symptoms, Sample Size) | Concussion Definition | Gait Outcome(s), Time Point(s), and Task(s) | Gait Outcome(s) Reported as Significantly Different With Concussion | Gait Outcome(s) Reported as Not Significantly Different With Concussion | DB |
|---|---|---|---|---|---|---|
| Repeated measures | Age: CONT: (20); CONC: (20) | | Time point(s): 1. Acute (CONC-acute), average days = 5.3; 2. After return to play (CONC-RTP), average days = 26.4 | | | |
| Canada | Male: 18 | | Task(s): 1. Walking in 5 possible directions (degrees) | | | |
| | n = 18 (9 concussions) | | | | | |
| | | | | | | |
| Buckley et al[26] | Intercollegiate athletes | CISG[2] | Outcome(s): 1. GV (m/s); 2. Propulsive force percentage (N); 3. Breaking force percentage (N) | 1. Decreased GV at day 1 (CI 95%, 1.1-1.2, $P = 0.01$, effect size = 1.14); 2. Increased propulsive force %: at day 1 (CI 95%, 0.24-0.64, $P < 0.010$, effect size = 0.85), and at day 10 (CI 95%, 0.23-0.69, $P < 0.01$, effect size = 0.85); 3. Increased breaking force % at day 1 (CI 95%, −0.17 to 0.07, $P < 0.010$, effect size = 0.540), and day 10 (CI 95%, 0.03-0.29, $P < 0.010$, effect size = 0.950) | None | 11 |
| Repeated measures | Mean age (SD; range): CONT: 20 (1.7; 19-22); CONC: 19 (1.0; 18-20) | | Time point(s): day 1 and 10 postconcussion | | | |
| USA | Sex not reported | | Task(s): 1. Self- selected walking | | | |
| | n = 47 (21 concussions) | | | | | |
| | | | | | | |
| Chiu et al[27] | Athletes | AAN[7] | Outcome(s): 1. GV(m/s) | 1. Decreased GV during Obstacle crossing ($P = 0.015$) and Walking + Q & A ($P = 0.040$) | None | 10 |
| No repeated measures | Mean age (SD; range): CONT: 21 (3.2; 18-25); CONC: 22 (3.6; 18-25) | | Time point(s): 48 h post concussion | | | |
| USA | Female: 18; Male: 28 | | Task(s): 1. Walking; 2. Walking + crossing obstacles; 3. Walking + Q & A | | | |
| | n = 46 (23 concussions) | | | | | |
| | | | | | | |
| Howell et al[28] | High school students | CISG[2] | Outcome(s): 1. Average GV (m/s); 2. Step length (m); 3. Step width (m); 4. Peak A–P COMV (m/s); 5. Peak M–L COMV (m/s); 6. Total M–L COMD (m) | 1. Decreased GV during Walk + MAS ($F_{1,37} = 6.02$, $P = 0.019$); 4. Decreased peak A–P COMV during Walk + MAS ($F_{1,37} = 6.230$, $P = 0.017$); 5. Increased peak M–L COMV ($F_{1,37} = 5.320$, $P = 0.027$); 6.Increased total M–L COMD during Walk + MAS ($F_{1,37} = 6.750$, $P = 0.013$) | 2. Step length; 3. Step width | 13 |
| Repeated measures | Mean age (SD; range): CONT: 16 (1; 14-17); CONC: 15 (1; 14-18) | | Time point(s): 72 h; 1, 2, 4, 8 wks postconcussion | | | |



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**TABLE 2. Summary of Data Extraction** *(Continued)*

| Study (Year, Design Follow-Up, Country) | Participants (Description, Age, Sex, Symptoms, Sample Size) | Concussion Definition | Gait Outcome(s), Time Point(s), and Task(s) | Gait Outcome(s) Reported as Significantly Different With Concussion | Gait Outcome(s) Reported as Not Significantly Different With Concussion | DB |
|---|---|---|---|---|---|---|
| USA | Female: 4; Male: 36 | | Task(s): 1. Walk; 2. Walk + MAS | | | |
| | n = 40 (20 concussions) | | | | | |
| | | | | | | |
| Kleffelgaard et al[29] | Adults | ACRM[7] | Outcome(s): 1. Normal walking speed (s); 2. Maximum walking speed (s); 3. 6 min Walk Test (m) | 1. Increased normal walking speed in CONC + self-reported balance problems ($P = 0.030$); 2. Increased maximum walking speed in CONC + self-reported balance problems ($P = 0.001$); 3. Decreased 6 min walking test among in CONC + self-reported balance problems ($P = 0.030$) | None | 14 |
| No repeated measures | Mean age (SD; range): CONC: 40 (12.8; 27-53); CONT: Baseline data | | Time point (s): Four years postconcussion | | | |
| Norway | Female: 10; Male: 19 | | Task(s): 1. CONC with and without self-reported balance problems | | | |
| | n = 29 (29 concussed) | | | | | |
| | | | | | | |
| Catena et al[30] | Young adults | AAN[7] | Outcome(s): 1. Peak sagittal COMV; 2. Peak frontal COMV; 3. Sagittal COP angular ROM (degrees); 4. Frontal COP angular ROM (degrees); 5. COM-COPS; 6. Frontal COMD (degrees) | 1. Decreased peak sagittal COMV at day 2 during Walk + SAS ($P = 0.064$); 3. Decreased sagittal COP angular ROM at day 2 during Walk + SAS ($P = 0.041$); 4. Increased frontal COP angular ROM at day 14 Walk + SAS ($P = 0.006$); 5. Decreased COM-COPS at day 2 ($P = 0.041$); 6. Increased frontal COMD at day 14 during Walk + SAS ($P = 0.006$) | 2. Peak frontal COMV | 11 |
| Repeated measures | Mean age (SD; range): CONT: 21 (4.1; 17-25); CONC: 21 (3.1; 18-24) | | Time point(s): 2, 6, 14, 28 d postconcussion | | | |
| USA | Female: 10; Male: 10 | | Condition Task(s): 1. Walking; 2. Walking + SAS | | | |
| | n = 20 (10 concussions) | | | | | |
| | | | | | | |
| Catena et al[31] | Collegiate athletes and non- athletes | AAN[7] | Outcome(s): 1. M–L COMV (m/s); 2. A–P COMV (m/s); 3. M–L COMD (m); 4. A–P COMD (m); 5. COM-COPS (m) | 1. Decreased M–L COMV at day 14 with short obstacle ($P = 0.014$), and day 28 with tall obstacle crossing ($P = 0.013$); 2. Decreased A–P COMV at day 2 during Walk + Q & A ($P = 0.014$); 3. Decreased M–L COMD at day 28 with short obstacle ($P = 0.000$), and tall obstacle crossing ($P = 0.001$); 4. Decreased A–P COMD at day 2 with Walk + Q & A ($P = 0.014$); 5. Decreased sagittal COM-COPS at day 2 during Walk + Q & A ($P = 0.038$) | None | 12 |



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**TABLE 2. Summary of Data Extraction** *(Continued)*

| Study (Year, Design Follow-Up, Country) | Participants (Description, Age, Sex, Symptoms, Sample Size) | Concussion Definition | Gait Outcome(s), Time Point(s), and Task(s) | Gait Outcome(s) Reported as Significantly Different With Concussion | Gait Outcome(s) Reported as Not Significantly Different With Concussion | DB |
|---|---|---|---|---|---|---|
| Repeated measures | Mean age (SD; range): CONT: 22 (3.1; 18-25); CONC: 22 (3.3; 19-25) | | Time point(s): 2, 6, 14, 28 d postconcussion | | | |
| USA | Female: 28; Male: 32 | | Task(s): 1. Walk; 2. Walk + short obstacle crossing; 3. Walk + tall obstacle crossing; 4. Walk + Q & A | | | |
| | n = 60 (30 concussions) | | | | | |
| | | | | | | |
| Catena et al[32] | Collegiate athletes | AAN[7] | Outcome(s): 1. Lead foot clearance (cm); 2. Trailing foot clearance (cm); 3. Obstacle contact (count) | 1. Increased lead foot crossing height variability at day 28 ($P = 0.012$); 3. Increased obstacle contact ($P = 0.003$) | 2. Trailing foot clearance | 11 |
| Repeated measures | Mean age (SD; range): CONT: 21 (2.3; 18-27); CONC: 21 (1.7; 18-24) | | Time point(s): 2, 6, 14, 28 d postconcussion | | | |
| USA | Female: 16; Male: 18 | | Task(s): 1. Walk + obstacle crossing; 2. Walk + obstacle crossing + Q & A | | | |
| | n = 34 (17 concussions) | | | | | |
| | | | | | | |
| Parker et al[33] | Collegiate athletes and nonathletes | AAN[7] | Outcome(s): 1. M–L COMD (m); 2. Peak M–L COMV (m/s); 3. Average GV (m/s); 4. Maximum A–P COM-COPS (m) | 1. Increased M–L COMD during Walk + mental tasks ($P = 0.002$); 2. Increased peak M–L COMV during Walk + mental task ($P = 0.001$); 3. Decreased GV during Walk + mental task ($P = 0.002$); 4. Decreased maximum A–P COM-COPS at days 2, 5, 14 with Walk + mental task | None | 12 |
| Repeated measures | Mean age (range): CONT: 22 (19-27); CONC: 22 (18-27) | | Time point(s): 2, 5, 14, 28 d postconcussion | | | |
| USA | Sex not reported | | Task(s): 1. Walk; 2. Walk + mental task (spelling, subtraction, and reciting backwards) | | | |
| | n = 56 (28 concussions) | | | | | |
| | | | | | | |
| Catena et al[34] | College students | AAN[7] | Outcome(s): 1. GV (m/s); 2. Stride length (m); 3. Average step width (m); 4. Stride time (s); 5. A–P COMD (m); 6. M–L COMD (m); 7. Peak M–L COMV (m/s); 8. Peak A–P COMV (m/s); 9. Maximum A–P COM-COPS (m); 10. Maximum M–L COM-COPS (m) | 1. Decreased GV ($P = 0.007$); 4. Increased stride time with Walk and Walk + Q & A ($P = 0.020$); 6. Increased M–L COMD with Walk and Walk + Q & A ($P = 0.041$); 7. Increased Peak M–L COMV with Walk and Walk + Q & A ($P = 0.046$); 8. Decreased peak A–P COMV ($P = 0.007$) | 2. Stride length; 3. Step width; 5. A–P COMD; 9. A–P COM-COPS; 10. M–L COM-COPS | 10 |
| No repeated measures | Mean age (SD; range): CONT: 22 (3; 19-25); CONC: 22 (4.5; 18-27) | | Time point(s): 48 h postconcussion | | | |
| USA | Female: 12; Male: 16 | | Task(s): 1. Walk; 2. Walk + Q & A; 3. Walk + SAS | | | |
| | n = 28 (14 concussions) | | | | | |
| | | | | | | |



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**TABLE 2. Summary of Data Extraction** *(Continued)*

| Study (Year, Design Follow-Up, Country) | Participants (Description, Age, Sex, Symptoms, Sample Size) | Concussion Definition | Gait Outcome(s), Time Point(s), and Task(s) | Gait Outcome(s) Reported as Significantly Different With Concussion | Gait Outcome(s) Reported as Not Significantly Different With Concussion | DB |
|---|---|---|---|---|---|---|
| Catena et al[35] | College students | AAN[7] | Outcome(s): 1. A–P COMD (m); 2. M–L COMD (m); 3. Peak A–P COMV (m/s); 4. Peak M–L COMV (m/s); 5. Maximum A–P COM-COPS (m); 6. Maximum M–L COM-COPS (m); 7. Instantaneous A–P velocity (m/s); 8. Instantaneous M–L velocity (m/s); 9. GV (m/s); 10. Stride length (m); 11. Step width (m); 12. Stride time (s) | 2. Increased M–L COMD with Walk + Q & A ($P = 0.045$); 3. Decreased peak A–P velocity ($P = 0.007$); 4. Increased peak M–L COMV ($P = 0.034$); 5. Decreased maximum A–P COM-COPS during Walk + Q & A task ($P = 0.038$); 7. Decreased Instantaneous A–P velocity with Walk + Q & A and obstacle crossing ($P = 0.010$); 9. Decreased GV ($P = 0.003$); 11. Increased step width with obstacle crossing ($P = 0.040$); 12. Increased stride time with Walk + Q & A and obstacle crossing ($P = 0.006$) | 1. A–P COMD; 6. Maximum M–L COM COPS (m/s); 8. Instantaneous M–L velocity; 10. Stride length | 10 |
| No repeated measures | Mean age (SD; range): CONT: 22 (3.1; 19-25); CONC: 22 (4.5; 18-27) | | Time point(s): A postconcussion time point | | | |
| USA | Female: 12; Male: 16 | | Task(s): 1. Walk; 2. Walk + obstacle crossing; 3. Walk + Q & A | | | |
| | n = 28 (14 concussions) | | | | | |
| | | | | | | |
| Parker et al[36] | Collegiate athletes | AAN[7] | Outcome(s): 1. M–L COMD; 2. Peak M–L COMV (m/s); 3. Average GV (m/s); 4. Maximum A–P; COM-COPS (m) | 1. Increased M–L COMD at days 5 and 28 with Walk + mental task ($P < 0.050$); 4. Increased maximum A–P COM-COPS ($P = 0.001$) | 2. Peak M–L COMV; 3. GV | 15 |
| Repeated measures | Mean age (SD; range): CONT: 21 (3.4; 18-25); CONC: 22 (3.3; 18-25) | | Time point(s): 2, 5, 14, 28 d postconcussion | | | |
| USA | Female: 28; Male: 30 | | Task(s): 1. Walk; 2. Walk + mental task; (spelling, subtraction, and reciting backwards) | | | |
| | n = 58 (29 concussions) | | | | | |
| | | | | | | |
| Parker et al[37] | College athletes | AAN[7] | Outcome(s): 1. GV (m/s); 2. Stride length (m); 3. Stride time (s); 4. Step width (m); 5. M–L COMD (m); 6. Anterior COMD (m); 7. Maximum M–L COM-COPS; 8. Maximum A–P COM-COPS; 9. Peak maximum instantaneous linear anterior COMV (m/s); 10. Peak maximum instantaneous linear M–L COMV (m/s) | 1. Decreased GV at day 2 ($P < 0.012$); 2. Decreased stride length at days 2 ($P < 0.016$); 5. Increased M–L COMD at days 2, 5, 28 ($P < 0.013$); 8. Decreased maximum A–P COM-COPS at day 2 with Walk task ($P < 0.005$); 8. Decreased maximum A–P COM-COPS at days 2, 14, 28 with Walk + mental tasks ($P < 0.005$) | 3. Stride time; 4. Step width; 6. Anterior COMD; 7. Maximum M–L COM-COPS; 9. Peak maximum instantaneous linear anterior COMV; 10. Peak maximum instantaneous linear M–L COMV | 13 |
| Repeated measures | Mean age (SD; range): CONT: 21 (1.8; 19-22); CONC: 21 (1.6; 19-22) | | Time point(s): 2, 5, 14, 28 d postconcussion | | | |
| USA | Female: 12; Male: 18 | | Task(s): 1. Walk; 2. Walk + mental task (spelling, subtraction, and reciting backwards) | | | |
| | n = 30 (15 concussions) | | | | | |



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**TABLE 2. Summary of Data Extraction** *(Continued)*

| Study (Year, Design Follow-Up, Country) | Participants (Description, Age, Sex, Symptoms, Sample Size) | Concussion Definition | Gait Outcome(s), Time Point(s), and Task(s) | Gait Outcome(s) Reported as Significantly Different With Concussion | Gait Outcome(s) Reported as Not Significantly Different With Concussion | DB |
|---|---|---|---|---|---|---|
| | | | | | | |
| Parker et al[38] | Collegiate athletes | AAN[7] | Outcome(s): 1. GV (m/s); 2. Step width (m); 3. Stride length (m); 4. Stride time (s); 5. A–P COMD (m); 6. M–L COMD (m); 7. Maximum instantaneous linear A–P COMV; 8. Maximum instantaneous linear M–L COMV; 9. Maximum A–P COM-COPS (m); 10. Maximum M–L COM-COPS (m) | 3. Decreased stride length ($P = 0.042$); 6. Increased M–L COMD with Walk + mental task, and decreased with Walk task ($P = 0.021$); 7. Decreased maximum instantaneous linear A–P COMV ($P = 0.041$) | 1. GV; 2. Step width; 4. Stride time; 5. A–P COMD; 8. Maximum instantaneous linear M–L COMV; 9. Maximum A–P COM-COPS (m); 10. Maximum M–L COM-COPS (m) | 10 |
| No repeated measures | Mean age (SD; range): CONT: 20 (1.9; 18-22); CONC: 20 (1.7; 19-22) | | Time point(s): Within 48 h postconcussion | | | |
| USA | Female: 12; Male: 8 | | Task(s): 1. Walk; 2. Walk + mental task (spelling, subtraction, and reciting backwards) | | | |
| | n = 20 (10 concussions) | | | | | |

*A–P, anterior to posterior; COM, center of mass; COMV, center of mass velocity; CONC, concussed; CONT, control; COP, center of pressure; LOC, loss of consciousness; m, meter; MAS, multiple auditory stimulus; M–L, medial to lateral; m/s, meter/second; Q & A, question and answers; RTP, return to practice; SAS, single auditory stimulus; SD, standard deviation.*

studies included: a limited description of the principal confounders (eg, concussion modifiers), insufficient information upon which to determine how the study sample was representative of the population of interest (ie, how the individuals who chose to participate differ from those who did not), inadequate sample size, and insufficient description of the validity and reliability of the measurement systems employed.

### *Synthesis of Results*

Table 3 summarizes the quantity, quality and level of evidence of the most frequently investigated gait parameters within the typical 10-day postconcussion recovery period, beyond the typical 10-day postconcussion recovery period, and after return to activity. Within the typical 10-day postconcussion recovery period there is a moderate amount of consistent level 4 evidence of increased M–L COMD; a moderate amount of inconsistent level 4 evidence of decreased GV, A–P COMV, and disturbed (ie, increased or decreased) M–L COMV and COM-COPS; and a small amount of inconsistent level 4 evidence of step and stride parameter alterations. Further, there is preliminary level 4 evidence that suggests that some of these gait deviations (ie, decreased GV and increased M–L COMD and M–L COMV) exist, and therefore may persist, beyond the typical 10-day postconcussion recovery period, and after return to activity, sport, recreation, or work.[21,23,29]

## DISCUSSION

To our knowledge, this is the first systematic review to identify and summarize quantifiable gait deviations associated with concussion that has incorporated both a formal evaluation of study quality and level of evidence. Although there is a lack of consensus about the most important gait parameters to assess after concussion, the current results suggest that concussed individuals sway more in the frontal plane (consistent level 4 evidence), and may walk slower (inconsistent level 4 evidence) compared with healthy controls within the typical 10-day postconcussion recovery period. Further, there is limited preliminary level 4 evidence that for some individuals, these deficits exist, and therefore may persist, beyond the typical 10-day postconcussion recovery period and return to activity. No studies were identified that assessed gait parameters to inform return to activity, sport, recreation, or work decisions. Further, few studies have assessed the association concussion and gait parameters in nonathletic populations.

It is important to highlight that the findings of this review are based upon an evaluation and synthesis of the existing literature and are limited by the design of the studies included. Overall there was a lack of high-quality evidence. The biggest threats to internal validity identified were related to selection bias, and the reporting and adjustment for potential confounding by factors such as preexisting gait deviations, the heterogenous nature of concussion (eg, not all individuals who suffer a concussion may develop gait deviations), presence or absence of symptoms, medication use, sleep disorder, and style of play. Similarly, there was potential for measurement bias across studies due to insufficient operationalization of many gait parameters and a lack of information about the measurement properties (eg, validity, reliability, resolution) of the measurement systems employed. In addition to limiting internal validity, the inability to assess for selection bias limits the degree to which the results of these studies can be generalized to the larger population from which the samples were drawn (external validity). The external validity of the results is brought further into question by the fact that 18



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**TABLE 3. Summary of Significant and Non-significant Gait Parameters by Quantity and Quality**

| Time Since Concussion | Gait Parameters | Studies With Repeated Measures | | Studies Without Repeated Measures | | Total Studies |
|---|---|---|---|---|---|---|
| | | SIG | NOT | SIG | NOT | |
| ≤10 d postconcussion | GV↓ | 5 (11-13) | 4 (11-15) | 4 (10) | 1 (10) | 14 |
| | M–L COMV* | 4 (12-16) | 5 (11-15) | 2 (10) | 1 (10) | 12 |
| | A–P COMV↓ | 4 (11-16) | 4 (11-13) | 3 (10) | — | 11 |
| | M–L COMD↑ | 6 (11-16) | — | 3 (10) | — | 9 |
| | COM-COPS* | 4 (11-13) | — | 1 (10) | 2 (10) | 7 |
| | STW | — | 3 (13-15) | 1 (10) | 2 (10) | 6 |
| | SRT | — | 1 (13) | 2 (10) | 1 (10) | 4 |
| | STL | 2 (12-13) | 1 (13) | — | 1 (10) | 4 |
| | SL | — | 1 (15) | 1 (10) | 1 (10) | 3 |
| >10 d postconcussion | M–L COMV* | 5 (122-16) | 3 (11-15) | — | 1 (11) | 9 |
| | M–L COMD↑ | 8 (11-16) | — | — | — | 8 |
| | GV↓ | 3 (12-13) | 2 (11-15) | 1 (14) | — | 6 |
| | A–P COMV↓ | 1 (13) | 2 (12-13) | — | — | 3 |
| | COM-COPS* | 3 (12-15) | — | — | — | 3 |
| After RTA | GV↓ | — | — | 1 (14) | — | 1 |
| | M–L COMD↑ | 1 (13) | — | 1 (11) | — | 2 |
| | M–L COMV↑ | — | — | 1 (11) | — | 1 |

*Cell values represent the number of studies (range of Downs and Black quality assessment tool scores).*
** Inconsistently reported direction of effect.*
*↑, consistently reported as increased; ↓, consistently reported as decreased; A–P COMV, anterior to posterior center of mass velocity; COM-COPS, center of mass – center of pressure separation; M–L COMD, medial to lateral center of mass displacement; M–L COMV, medial to lateral center of mass velocity; NOT, studies that do not report a statistically significant difference between concussed and nonconcussed study groups; RTA, return to activity; SIG, studies that report a statistically significant difference between concussed and nonconcussed study groups.*

(86%) of the studies included athletes or those who were involved in athletic-like activities. Further, as 15 (71%) included studies had sample sizes less than 50 participants (inclusive of concussed and healthy controls), and only 1 study had a sample size greater than 100,[20] it is possible that some of the included studies were inadequately powered to detect effects and overestimate the reported effect sizes.[39]

Another consideration is the lack of information justifying the time points chosen for postconcussion follow-up gait testing. To improve the clinical utility of future studies it is suggested that investigators consider a follow-up gait-testing schedule based on commonly accepted concussion recovery stages. For instance, the CIGS, report that 80% to 90% of individuals who sustain a concussion recover within 7 to 10 days.[2] Further, it is recommended that concussed individuals should progress through a Graded Return to Physical Activities Protocol (GRTPP) before being fully cleared to participate in physical activities (Full Clearance). This approach will improve our understanding of gait during each of the concussion recovery stages, and the utility of gait analysis as a concussion recovery measure.

The findings of the current review build upon a previous meta-analyses that examined the utility of a dual-task paradigm (DT) for sports-related concussion gait assessment,[40] that reported decreased GV [pooled mean difference (95% CI); −0.133 m/s (−0.197 to −0.069)] and greater M–L COMD [0.007 m (0.002-0.011)] 2 days postconcussion, and decreased M–L COMV at 6 [0.014 m/s, (0.003-0.026)] and 28 [0.013 m/s (0.003-0.023)] days postconcussion. Taken together, the findings of this and the previous review suggest that concussed individuals may initially adopt a conservative approach to gait which involves walking slower and keeping their COM closer to their base of support (ie, COP) which increases their M–L sway. This gait pattern is similar to that reported amongst other populations with a high risk of falling (eg, moderate to severe traumatic brain injuries and elderly).[41,42] These findings have implications for clinical tests designed to assess and detect gait disturbances in individuals who have suffered a concussion. Specifically, a clinical test should include an assessment of GV and M–L sway to be sensitive to gait disturbances that may occur after concussion. The most common clinical test of gait in concussed patients is the Tandem Gait Test (TGT).[43] This test involves walking in a forward direction as quickly and as accurately as possible along a 38-mm wide, three-meter line and back, with an alternate foot heel-to-toe gait. The total time it takes to complete the task, the ability to stay on the line, and avoid separation of their heel and toe while maintaining balance are noted. Although rudimentary, the TGT does include a component of GV (distance × time) and M–L sway (ability to walk on a line). With that said, the diagnostic accuracy of the TGT for detecting gait disturbance is unknown, as concussed athletes have been shown to complete the test faster than non-concussed athletes.[44]

A novel finding of this review is that concussion-associated gait deficits, including decreased GV and increased M–L COMD and M–L COMV exist, and therefore may possibly persist beyond the typical 10-day recovery period after concussion, and after return to activity, sport, recreation, and work. The persistence of gait disturbances beyond return to activity, sport, and recreation may place an athlete at



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increased risk of future injury. This hypothesis is supported by initial evidence from professional rugby that demonstrates a 60% higher incidence of injury in players after concussion [incidence ratio rate (95% CI, 1.6 [1.4-1.8])] compared with players who did not suffer a concussion.[45] Further, that, the median time to injury after return to sport was shorter among players who suffered a concussion [53 days (95% CI, 41-46)] than players who did not [114 days (95% CI, 85-143)]. Further investigation of the relationship between concussion-related gait deficits and injury risk in the postconcussion return to activity, sport, recreation, and work period is required.

### *Limitations*

Meta-analyses were not possible due to the heterogeneity of gait parameters assessed, and variable timing of postconcussion gait analysis. Despite a comprehensive search strategy, and the rigorous approach to study selection and data extraction, it is important to acknowledge the possibility of omitting a relevant study and inclusion of only Arabic and English language studies as additional potential limitations. As the findings of this review are based on the existing literature, it is important to consider that not all possibly relevant gait parameters may have been considered. Further, it is important to highlight that the associations between concussion and various gait parameters identified in this review are based on level 4 evidence with a high risk of bias, and given that 76% of the included studies were performed by 3 research teams the findings may lack external validity (ie, generalizability). Finally, it is important to acknowledge that the findings related to the existence of gait deviations after the typical 10-day postconcussion recovery period and return to activity are based on a smaller number of investigations.

### *Recommendations*

There is a need for high-quality prospective studies with sufficient sample sizes spanning the preinjury through to concussion and return to activity, sport, recreation, and/or work, and beyond. This research should use a definition for concussion that is consistent with the AAN, ACRM, CISG, or NATA criteria, and consider quantifying gait parameters that appear to be the most useful for detecting gait deficits postconcussion (ie, GV and M–L COMD) within the first 10 days after concussion and at the time of symptom resolution, the start of graded activity or GRTPP, and at full clearance to return to activity, sport, recreation, and/or work. Further, there is a need for studies examining the association between concussion and gait across nonathletic populations. Finally, from a clinical perspective, there is a need for a dynamic balance assessment tool that is capable of detecting gait disturbances in concussed individuals in field settings. Ideally, this tool would challenge an individual's GV and M–L COMD under complex physical tasks (ie, walking, sport, and/or work-specific tasks) with and without secondary cognitive demands to better understand alterations that may occur.

## CONCLUSIONS

Individuals who have suffered a concussion may sway more in the frontal plane (consistent level 4 evidence), and walk slower (inconsistent level 4 evidence) compared to healthy controls within the typical 10-day postconcussion recovery period. Further, there is preliminary level 4 evidence that for some individuals, these deficits exist, and therefore may possibly persist, beyond the typical 10-day postconcussion recovery period and return to activity. There is a paucity of information about the role that gait parameters might have in informing return to activity, sport, recreation, or work decisions, and the current level of evidence is threatened by a risk of bias. Future research should include high-quality prospective studies spanning the period from concussion through return to activity and beyond to better understand the natural course of gait alterations after concussion across diverse populations.

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## APPENDIX 1. SEARCH STRATEGIES AND RESULTS

### TABLE 1. Ovid Medline Search Strategy

| | |
|---|---|
| Provider/Interface | Ovid |
| Database | Medline |
| Date searched | December 14, 2015. Updated September 29, 2016 |
| Database update | — |
| Search developer(s) | TM & LD |
| Limit to English? | No |
| Date Range | 1974–2016 |

| | |
|---|---|
| 1 | exp brain hemorrhage, traumatic/ |
| 2 | [concuss* or postconcuss* or acquired brain damage or acquired brain injur* or mTBI or traumatic brain injur* or minor brain injur* or minor head injur* or mild head injur* or commotio cerebri].mp. |
| 3 | [[brain injur* or brain damage or head injur*] adj12 [trauma* or accident* or hit or blow or fall*]].mp. |
| 4 | [gait or walk or walking or locomotion or ambulation or stride length or stride width or step length or step width or cadence or [[centre of mass or center or mass or range of motion] adj4 [displace* or sway*]]].mp. |
| 5 | (1 or 2 or 3) and 4 |
| 6 | limit 5 to animals |
| 7 | limit 6 to humans |
| 8 | 5 not (6 not 7) |
| 9 | limit 8 to “review articles” |
| 10 | 8 not 9 |
| 11 | [cadaver* or mouse or mice or rat or rats or rabbit* or pig or pigs or porcine or murine or sheep].ti. |
| 12 | 10 not 11 |
| 13 | case reports/ |
| 14 | 12 not 13 |

### TABLE 2. SCOPUS search strategy

| | |
|---|---|
| Provider/Interface | |
| Database | SCOPUS |
| Date searched | December 14, 2015. Updated September 29, 2016 |
| Database update | — |
| Search developer(s) | TM & LD |
| English only? | NO |
| Date Range | 1974–2016 |

| | |
|---|---|
| 1 | TITLE-ABS-KEY [ concuss* OR postconcuss* OR “acquired brain damage” OR “acquired brain injur*” OR mtbi OR “traumatic brain injur*” OR “minor brain injur*” OR “minor head injur*”OR “mild head injur*” OR “ommotion cerebri” ] AND TITLE-ABS-KEY [ gait OR walk OR walking OR locomotion OR ambulation OR “stride length” OR “stride width” OR “step length” OR “step width” OR cadence OR “center of mass displacement” OR sway ] ] AND [ EXCLUDE [ DOCTYPE, “re” ] OR EXCLUDE [ DOCTYPE, “cp” ] OR EXCLUDE [ DOCTYPE, “cr” ] ] |



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**TABLE 3. Ovid PsycINFO Search Strategy**

| | |
|---|---|
| Provider/Interface | Ovid |
| Database | PsycINFO |
| Date searched | December 14, 2015. Updated September 29, 2016 |
| Database update | — |
| Search developer(s) TM & LD | |
| Limit to English? | No |
| Date Range | 1974–2016 |
| 1 | exp traumatic brain injury/ |
| 2 | [concuss* or postconcuss* or acquired brain damage or acquired brain injur* or mTBI or traumatic brain injur* or minor brain injur* or minor head injur* or mild head injur* or commotio cerebri].mp. |
| 3 | [[brain injur* or brain damage or head injur*] adj12 [trauma* or accident* or hit or blow or fall*]].mp. |
| 4 | [gait or walk or walking or locomotion or ambulation or stride length or stride width or step length or step width or cadence or [[centre of mass or center or mass or range of motion] adj4 [displace* or sway*]]].mp. |
| 5 | (1 or 2 or 3) and 4 |
| 6 | limit 5 to animal |
| 7 | limit 6 to human |
| 8 | 5 not (6 not 7) |
| 9 | limit 8 to reviews |
| 10 | case report/ |
| 11 | 8 not 10 |

**TABLE 4. CINAHL Search Strategy**

| | |
|---|---|
| Provider/Interface | Ebsco |
| Database | CINAHL |
| Date searched | December 14, 2015. Updated September 29, 2016 |
| Database update | — |
| Search developer(s) TM & LD | |
| Limit to English? | No |
| Date Range | 1974–2016 |
| S1 | [ [MH "Brain Concussion+"] OR [MH "Right Hemisphere Injuries"] ] OR [ [concuss* or postconcuss* or "acquired brain damage" or "acquired brain injur*" or mTBI or "traumatic brain injur*" or "minor brain injur*" or "minor head injur*" or "mild head injur*" or "commotio cerebri"] ] OR [ [["brain injur*" or "brain damage" or head "injur*"] n12 [trauma* or accident* or hit or blow or fall*]] ] |
| S2 | gait or walk or walking or locomotion or ambulation or "stride length" or "stride width" or "step length" or "step width" or cadence or [["centre of mass" or "center of mass" or "range of motion"] n4 [displace* or sway*]]] |
| S3 | S1 AND S2 |
| S4 | PT case study or meta analysis or review |
| S5 | S3 NOT S4 |
| S6 | TI cadaver* or mouse or mice or rat or rats or rabbit* or pig or pigs or porcine or murine or sheep |
| S7 | S5 NOT S6 |



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**TABLE 5. Ovid EMBASE Search Strategy**

| | |
|---|---|
| Provider/Interface | Ovid |
| Database | EMBASE |
| Date searched | December 14, 2015. Updated September 29, 2016 |
| Database update | — |
| Search developer(s) TM & LD | |
| Limit to English? | No |
| Date Range | 1974–2016 |
| 1 | Traumatic brain injury/ |
| 2 | [concuss* or postconcuss* or acquired brain damage or acquired brain injur* or mTBI or traumatic brain injur* or minor brain injur* or minor head injur* or mild head injur* or commotio cerebri].mp. |
| 3 | [[brain injur* or brain damage or head injur*] adj12 [trauma* or accident* or hit or blow or fall*]].mp. |
| 4 | 1 or 2 or 3 |
| 5 | [gait or walk or walking or locomotion or ambulation or stride length or stride width or step length or step width or cadence or [[centre of mass or center or mass or range of motion] adj4 [displace* or sway*]]].mp. |
| 6 | 4 and 5 |
| 7 | limit 6 to animals |
| 8 | limit 7 to human |
| 9 | 6 not 7 |
| 10 | limit 9 to (conference abstract or "conference review" or "review") |
| 11 | 9 not 10 |
| 12 | case report/ |
| 13 | case study/ |
| 14 | 11 not (12 or 13) |
| 15 | [cadaver* or mouse or mice or rat or rats or rabbit* or pig or pigs or porcine or murine or sheep].ti. |
| 16 | 14 not 15 |

**TABLE 6. EBSCO SportDiscus Search Strategy**

| | |
|---|---|
| Provider/Interface | Ovid |
| Database | EMBASE |
| Date searched | December 14, 2015. Updated September 29, 2016 |
| Database update | — |
| Search developer(s) TM & LD | |
| Limit to English? | No |
| Date Range | 1974–2016 |
| S1 | [ [concuss* or postconcuss* or "acquired brain damage" or "acquired brain injur*" or mTBI or "traumatic brain injur*" or "minor brain injur*" or "minor head injur*" or "mild head injur*" or "commotio cerebri"] ] OR [ [["brain injur*" or "brain damage" or head "injur*"] n12 [trauma* or accident* or hit or blow or fall*]] |
| S2 | Gait or walk or walking or locomotion or ambulation or "stride length" or "stride width" or "step length" or "step width" or cadence or ["[centre of mass" or "center of mass" or "range of motion"] n4 [displace* or sway*]]] |
| S3 | S1 AND S2 |
| S4 | [ S1 AND S2 ] NOT TI [ cadaver* or mouse or mice or rat or rats or rabbit* or pig or pigs or porcine or murine or sheep ] |



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## APPENDIX 2. UNIQUE ARTICLE IDENTIFICATION BY DATABASE

| Database | Items Found | Internal Duplicates | External Duplicates | Unique Items |
|---|---|---|---|---|
| MEDLINE | 466 | 8 | 172 | 286 |
| PsycINFO | 217 | 0 | 25 | 192 |
| CINAHL | 244 | 1 | 0 | 243 |
| EMBASE | 552 | 13 | 303 | 236 |
| SCOPUS | 969 | 5 | 538 | 426 |
| Sport Discus | 202 | 2 | 120 | 80 |



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